

Т5. Случ. вел. $\xi \sim R(\theta, 2\theta)$, $p(x) = \frac{1}{\theta} \{(\theta, 2\theta)\}$

$$M\xi = \int_{\theta}^{2\theta} x \cdot \frac{1}{\theta} dx = \frac{1}{\theta} \frac{x^2}{2} \Big|_{\theta}^{2\theta} = \frac{1}{2\theta} (4\theta^2 - \theta^2) = \frac{3}{2}\theta$$

$$M\xi^2 = \int_{\theta}^{2\theta} x^2 \cdot \frac{1}{\theta} dx = \frac{1}{3\theta} (8\theta^3 - \theta^3) = \frac{7}{3}\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = \frac{7}{3}\theta^2 - \left(\frac{3}{2}\theta\right)^2 = \frac{28 - 27}{12}\theta^2 = \frac{\theta^2}{12}$$

а) ОММ: $\alpha_1(\theta) = \hat{\alpha}_1$, $\frac{3}{2}\hat{\alpha}_1 = \bar{X} \longrightarrow$
 $\longrightarrow \hat{\alpha}_1 = \frac{2}{3}\bar{X}$

ОМП: $L(\theta) = \prod_{i=1}^n p(x_i, \theta) = \frac{1}{\theta^n} \{(\theta \leq x_i \leq 2\theta)\}$

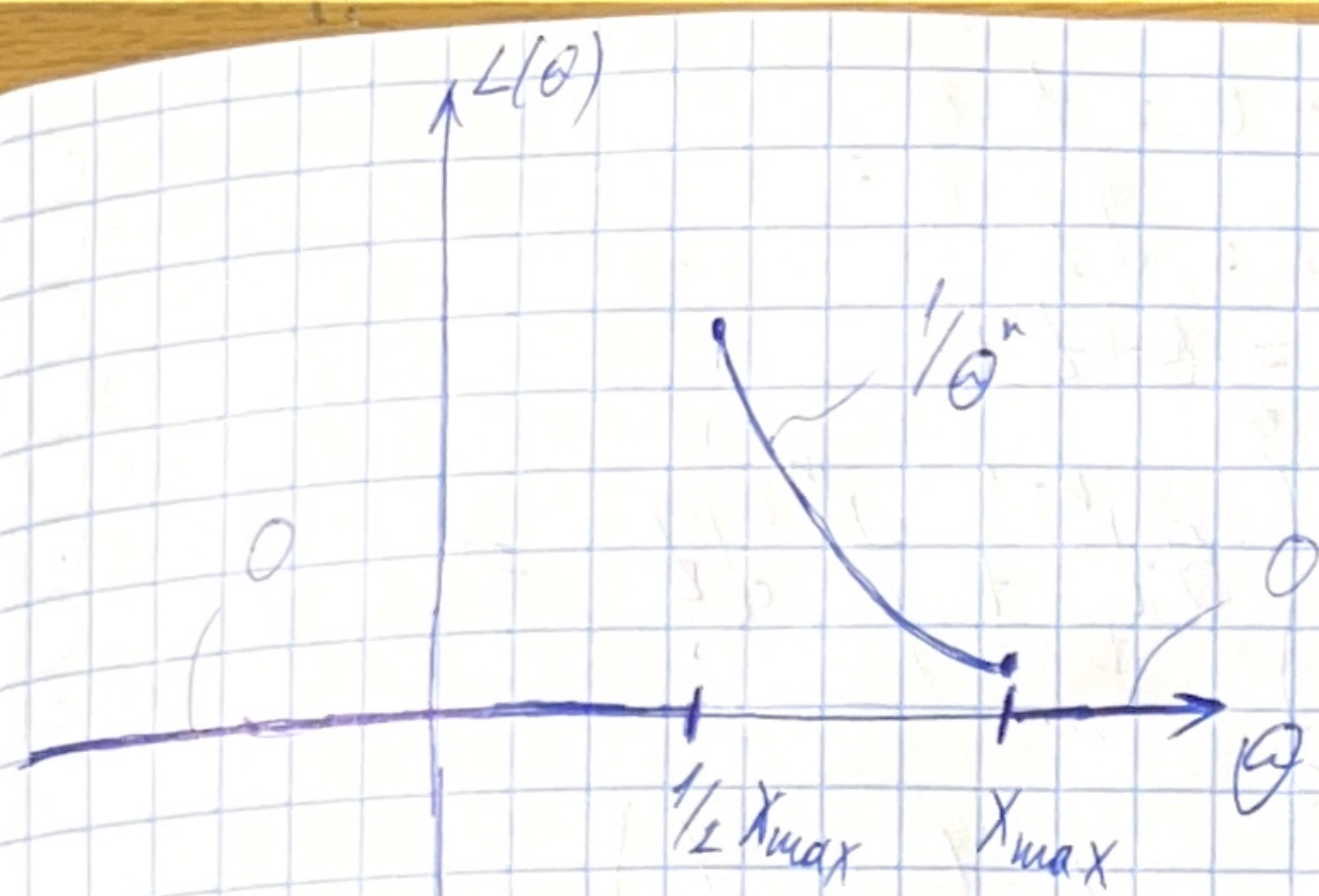
$$\ln L(\theta) = \ln \theta^{-n} = -n \ln \theta$$

$\left\{ \begin{array}{l} \text{м.к. отрезок } [\theta; 2\theta], \text{ то } \theta > 0, \text{ м.к. } \\ 2\theta > \theta, \theta > 0. \end{array} \right\}$

$(\ln L(\theta))' = -\frac{n}{\theta}$ - монот. убыв.

$X_{\max} = \max x_i \leq 2\theta \Rightarrow \theta \geq \frac{1}{2} X_{\max}$

$\max x_i \geq \theta \Rightarrow \theta \in [\frac{1}{2} X_{\max}, X_{\max}]$



$$\Rightarrow \hat{\theta}_2 = \frac{1}{2} X_{\max}$$

б) $\hat{\theta}_1$:

Немешённость: $M\hat{\theta}_1 = M\left(\frac{2}{3}\bar{X}\right) = \frac{2}{3}\mu_{\xi} =$
 $= \frac{2}{3} \cdot \frac{3}{2}\theta = \theta \Rightarrow$ немешу.

Согласительность: $\forall \theta > 0 \hat{\theta}_1$ - немешу и

$$D\hat{\theta}_1 = D\left(\frac{2}{3}\bar{X}\right) = \frac{4}{9} D\left(\frac{1}{n}\sum x_i\right) = \frac{4D\xi}{9n} =$$

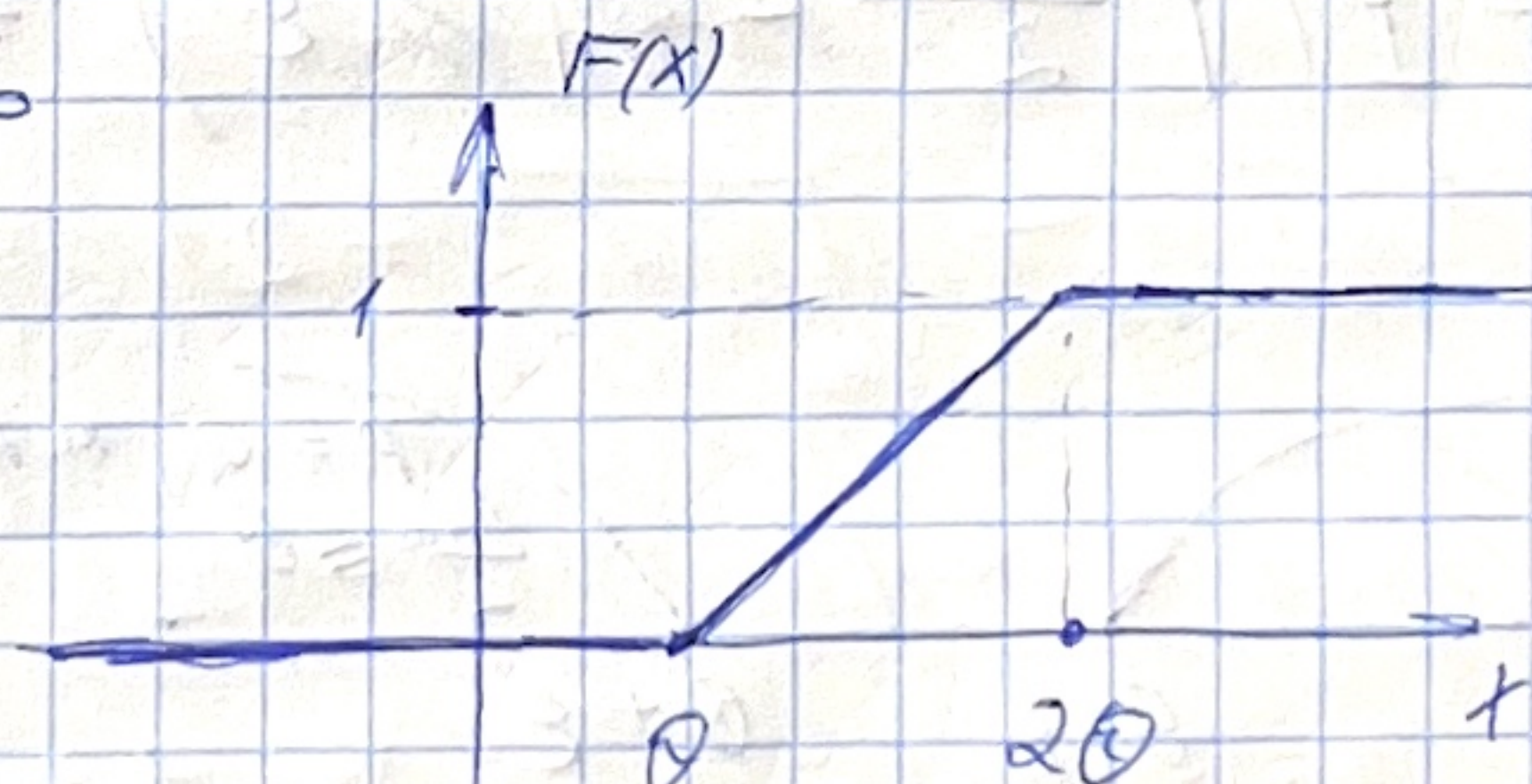
$$= \frac{4 \cdot \theta^2}{9n \cdot 12} = \frac{\theta^2}{27n} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \text{по раш. ул.}$$

оценка явл. согласительной

$\hat{\theta}_2$:

$\xi \sim R(\theta, 2\theta), F(x) = \int_{-\infty}^x p(t)dt \quad \ominus$

$$\ominus \begin{cases} 0, & x \leq \theta \\ \frac{x-\theta}{\theta}, & x \in (\theta; 2\theta) \\ 1, & x > 2\theta \end{cases}$$



$$X_{\max} \sim (F(z))^n = \psi(z); p(z) = \psi'(z) =$$

$$= \left(\frac{z-\theta}{\theta}\right)^{n-1} \cdot \{(\theta; 2\theta)\} = \frac{n}{\theta} \left(\frac{z-\theta}{\theta}\right)^{n-1} \cdot \{(\theta; 2\theta)\} =$$

$$= \frac{n}{\theta} \left(\frac{z-\theta}{\theta}\right)^{n-1} \cdot \{(\theta; 2\theta)\}$$

$$M_{X_{\max}} = \int_{\theta}^{2\theta} x \frac{n}{\theta} \left(\frac{x-\theta}{\theta}\right)^{n-1} dx = \frac{n}{\theta^n} \int_{\theta}^{2\theta} x(x-\theta)^{n-1} dx =$$

$$= \frac{n}{\theta^n} \int_{\theta}^{2\theta} x(x-\theta)^{n-1} dx = \left\{ \begin{array}{l} x-\theta = t \\ dx = dt \\ x = \theta+t \end{array} \right\} =$$

$$= \frac{n}{\theta^n} \int_0^{\theta} (\theta+t)t^{n-1} dt = \frac{n}{\theta^n} \int_0^{\theta} (\theta t^{n-1} + t^n) dt =$$

$$= \frac{n}{\theta^n} \left(\theta \cdot \frac{t^n}{n} + \frac{t^{n+1}}{n+1} \right) = n\theta \left(\frac{1}{n} + \frac{1}{n+1} \right)$$

$$M \tilde{\theta}_2 = M \left(\frac{1}{2} X_{\max} \right) = \frac{1}{2} M X_{\max} = \frac{1}{2} n\theta \left(\frac{1}{n} + \frac{1}{n+1} \right) =$$

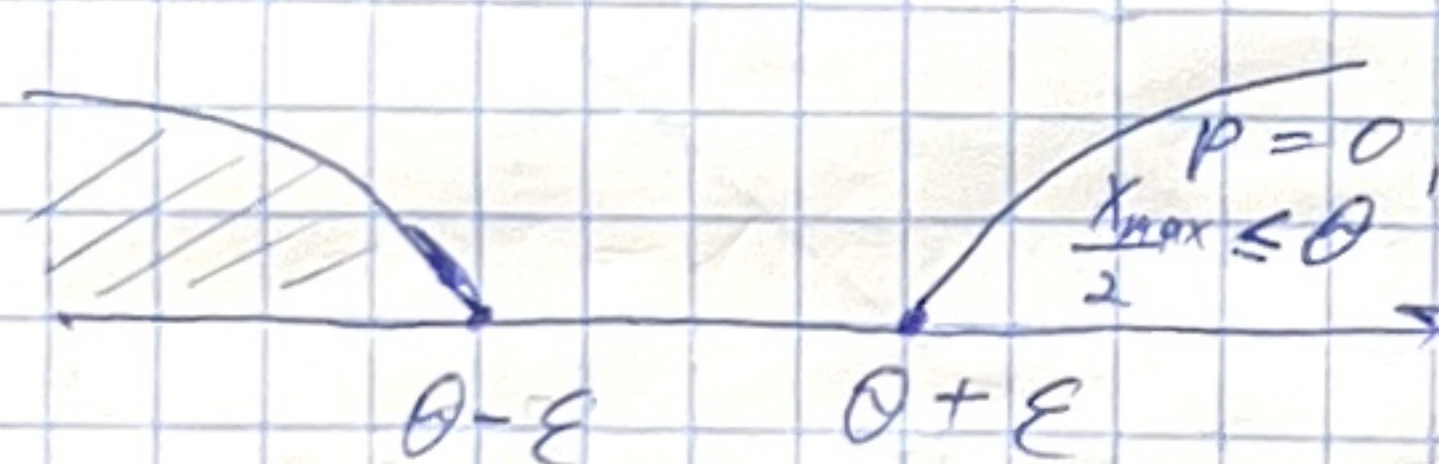
$$= \frac{\theta}{2} \left(1 + \frac{n}{n+1} \right) = \frac{\theta}{2} \left(\frac{2n+1}{n+1} \right) \neq \theta \Rightarrow \text{не явл. несмещ. (но явл. асимпт. несмещ.)}$$

Управляемая оценка: $\tilde{\tilde{\theta}}_2 = \frac{2(n+1)}{2n+1} \tilde{\theta}_2 =$

$$= \frac{2n+2}{2n+1} \tilde{\theta}_2$$

Состоятельность $\tilde{\tilde{\theta}}_2$: $P(|\tilde{\tilde{\theta}}_2 - \theta| \geq \varepsilon) =$

$$= P\left(\left| \frac{X_{\max}}{2} - \theta \right| \geq \varepsilon \right) = P\left(\frac{X_{\max}}{2} \leq \theta - \varepsilon \right) =$$

$$= P(X_{\max} \leq 2(\theta - \varepsilon))$$


или $2(\theta - \varepsilon) < \theta$, т.е. $\theta < 2\varepsilon$: $P(X_{\max} \leq 2(\theta - \varepsilon)) = 0 \xrightarrow{n \rightarrow \infty} 0$

($X_{\max} \in [\theta; 2\theta]$)

Иначе: $P(X_{\max} \leq 2(\theta - \varepsilon)) = \left(\frac{2(\theta - \varepsilon) - \theta}{\theta} \right)^n =$

$$= \left(\frac{\theta - 2\varepsilon}{\theta} \right)^n \text{ т.к. } \theta \geq 2\varepsilon, \text{ то } \theta - 2\varepsilon \geq 0, \text{ но}$$

$$\theta - 2\varepsilon < \theta \text{ (т.к. } \varepsilon > 0 \text{)}. \text{ т.е. } 0 \leq \frac{\theta - 2\varepsilon}{\theta} < 1 \Rightarrow$$

$$\Rightarrow \left(\frac{\theta - 2\varepsilon}{\theta} \right)^n \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \forall \varepsilon > 0 \forall \theta > 0 \quad P \xrightarrow{n \rightarrow \infty} 0$$

$\Rightarrow$ оценка явл. состоятельной

Считаем, что $\tilde{\theta}_2$ - меньше. и $\mathcal{D}\tilde{\theta}_2 = \mathcal{D}\left(\frac{2(n+1)}{2n+1}\tilde{\theta}_2\right) =$

$$= \frac{4(n+1)^2}{(2n+1)^2} \mathcal{D}\tilde{\theta}_2 = \frac{4(n+1)^2}{(2n+1)^2} \mathcal{D}\left(\frac{1}{2}X_{\max}\right) = \frac{(n+1)^2}{(2n+1)^2} \mathcal{D}X_{\max}$$

Вычисляем $\mathcal{D}X_{\max}$:

$$M_{X_{\max}}^2 = \int_{\theta}^{\theta+2\theta} X^2 \cdot \frac{n}{\theta} \left(\frac{X-\theta}{\theta}\right)^{n-1} dX = \frac{n}{\theta^n} \int_{\theta}^{\theta+2\theta} X^2 (X-\theta)^{n-1} dX =$$

$$\stackrel{X-\theta=t}{dx=dt} = \frac{n}{\theta^n} \int_0^{2\theta} (t+\theta)^2 t^{n-1} dt = \frac{n}{\theta^n} \int_0^{2\theta} (t^2 + 2\theta t + \theta^2) t^{n-1} dt =$$

$$= \frac{n}{\theta^n} \int_0^{2\theta} (t^{n+1} + 2\theta t^n + \theta^2 t^{n-1}) dt =$$

$$= \frac{n}{\theta^n} \left(\frac{\theta^{n+2}}{n+2} + \frac{2\theta \cdot \theta^{n+1}}{n+1} + \frac{\theta^2 \cdot \theta^n}{n} \right) = n\theta^2 \left(\frac{1}{n+2} + \frac{2}{n+1} + \frac{1}{n} \right)$$

$$\mathcal{D}X_{\max} = M_{X_{\max}}^2 - (M_{X_{\max}})^2 =$$

$$= \theta^2 \left(\frac{n}{n+2} + \frac{2n}{n+1} + 1 \right) - \theta^2 \left(1 + \frac{n}{n+1} \right)^2 =$$

$$= \theta^2 \left(\frac{n}{n+2} + \frac{2n}{n+1} + 1 \right) - \theta^2 \left(1 + \frac{2n}{n+1} + \frac{n^2}{(n+1)^2} \right) =$$

$$= \theta^2 \cdot \frac{n}{n+2} - \theta^2 \cdot \frac{n^2}{(n+1)^2} \xrightarrow{n \rightarrow \infty} \theta^2 - \theta^2 = 0$$

$$\rightarrow \mathcal{D}\tilde{\theta}_2 = \frac{(n+1)^2}{(2n+1)^2} \mathcal{D}X_{\max} \xrightarrow{n \rightarrow \infty} \frac{1}{4} \cdot 0 = 0$$

$\Rightarrow$ по формуле оценки для смещения

с) Сравним эффективность $\hat{\theta}_1$ и $\tilde{\theta}_2$

$$\mathcal{D}\hat{\theta}_1 = \frac{\theta^2}{27n}, \quad \mathcal{D}\tilde{\theta}_2 = \frac{(n+1)^2}{(2n+1)^2} \mathcal{D}X_{\max}$$

$$\frac{\theta^2}{27n} \vee \frac{(n+1)^2}{(2n+1)^2} \cdot \theta^2 \left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2} \right)$$

$$\frac{(n+1)^2}{(2n+1)^2} \left(\frac{n(n^2+2n+1) - n^2(n+2)}{(n+2)(n+1)^2} \right) = \frac{(n+1)^2}{(2n+1)^2} \left(\frac{n^3 + 2n^2 + n - n^3 - 2n^2}{(n+2)(n+1)^2} \right)$$

$$= \frac{n}{(n+2)(2n+1)^2}$$

$$\Rightarrow \frac{1}{27n} \vee \frac{n}{(n+2)(2n+1)^2}$$

$$27n^2 \vee (n+2)(4n^2+4n+1) = 4n^3 + 4n^2 + n + 8n^2 + 8n + 2$$

$$27n^2 \vee 4n^3 + 12n^2 + 9n + 2$$

$$0 \vee 4n^3 - 15n^2 + 9n + 2 = k(n)$$

Рассм. $(4x^3 - 15x^2 + 9x + 2)' = 12x^2 - 30x + 9$ при

$$x \geq 1 \quad D = 900 - 4 \cdot 9 \cdot 12 = 468$$

$$x_{1,2} = \frac{30 \pm \sqrt{468}}{24} \quad x_1 \approx 0,34861$$

$$x_2 \approx 2,15138$$

$$\sqrt{468} > \sqrt{441} = 21 \Rightarrow x_1 < \frac{30-21}{24} = \frac{9}{24} < 1$$

при $x=1$ $4 \cdot 1^3 - 15 \cdot 1^2 + 9 \cdot 1 + 2 = 0$

$$4 \cdot 3^3 - 15 \cdot 3^2 + 9 \cdot 3 + 2 = 2$$

$$4 \cdot 2^3 - 15 \cdot 2^2 + 9 \cdot 2 + 2 = -8 \Rightarrow 2 \leq 2\text{-й корень} \leq 3$$

1-ый корень = 1 при $x > x_2$ $\uparrow\uparrow$

Итак: при $1 \leq n \leq 2$: $4n^3 - 15n^2 + 9n + 2 < 0$

$$\Rightarrow \frac{1}{27n} < \frac{n}{(n+2)(2n+1)^2} \quad \text{и} \quad \hat{\theta}_1 \text{ эффективнее } \hat{\theta}_2$$

при $n \geq 3$: $\frac{1}{27n} > \frac{n}{(n+2)(2n+1)^2}$ и $\hat{\theta}_2$ эффективнее $\hat{\theta}_1$

d) В качестве $f(\bar{X}_n, \theta)$ возьмем $\frac{X_{\max} - \theta}{\theta}$

$$\Phi(t) = P\left(\frac{X_{\max} - \theta}{\theta} < t\right) = P(X_{\max} < \theta + \theta t) =$$

$$= \begin{cases} 0, & \theta t + \theta \leq \theta, \text{ т.е. } t \leq 0 \\ \left(\frac{\theta + \theta t - \theta}{\theta}\right)^n = t^n, & 0 < \theta + \theta t < 2\theta, \text{ т.е. } t \in (0; 1) \\ 1, & \theta t + \theta \geq 2\theta, \text{ т.е. } t \geq 1 \end{cases}$$

$$f(t) = \Phi'(t) = n t^{n-1} \quad \{ (0; 1) \}$$

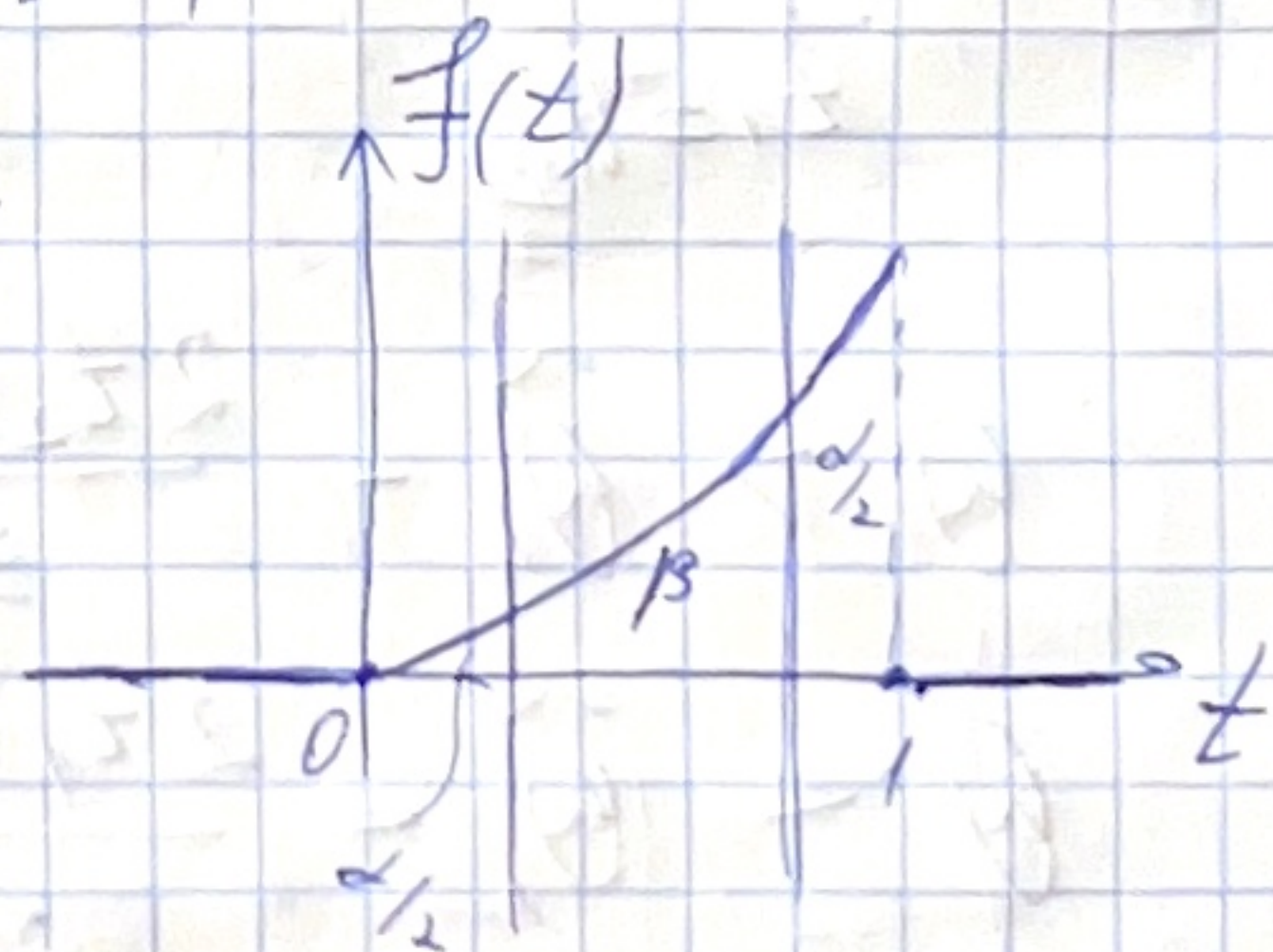
$$\Phi(X_p) = p = X_p^n \Rightarrow X_p = \sqrt[n]{p}$$

$$q_{\frac{1-\beta}{2}} = t_1 = \sqrt[n]{\frac{1-\beta}{2}}$$

$$q_{\frac{1+\beta}{2}} = t_2 = \sqrt[n]{\frac{1+\beta}{2}}$$

$$P(t_1 < f(\bar{X}_n, \theta) < t_2) = \beta$$

$$\sqrt[n]{\frac{1-\beta}{2}} < \frac{X_{\max} - \theta}{\theta} < \sqrt[n]{\frac{1+\beta}{2}} \Rightarrow 1 + t_1 < \frac{X_{\max}}{\theta} < 1 + t_2$$



$$\Rightarrow \frac{1}{1+t_1} > \frac{\theta}{X_{\max}} > \frac{1}{1+t_2} \Rightarrow$$

$$\Rightarrow \frac{X_{\max}}{1 + \sqrt[n]{\frac{1-\beta}{2}}} > \theta > \frac{X_{\max}}{1 + \sqrt[n]{\frac{1+\beta}{2}}}$$

$$\left[\frac{X_{\max}}{1 + \sqrt[n]{\frac{1+\beta}{2}}} < \theta < \frac{X_{\max}}{1 + \sqrt[n]{\frac{1-\beta}{2}}} \right] \quad \text{— Точный доверительный интервал}$$

е) Асимптотический доверительный интервал по ОММ:

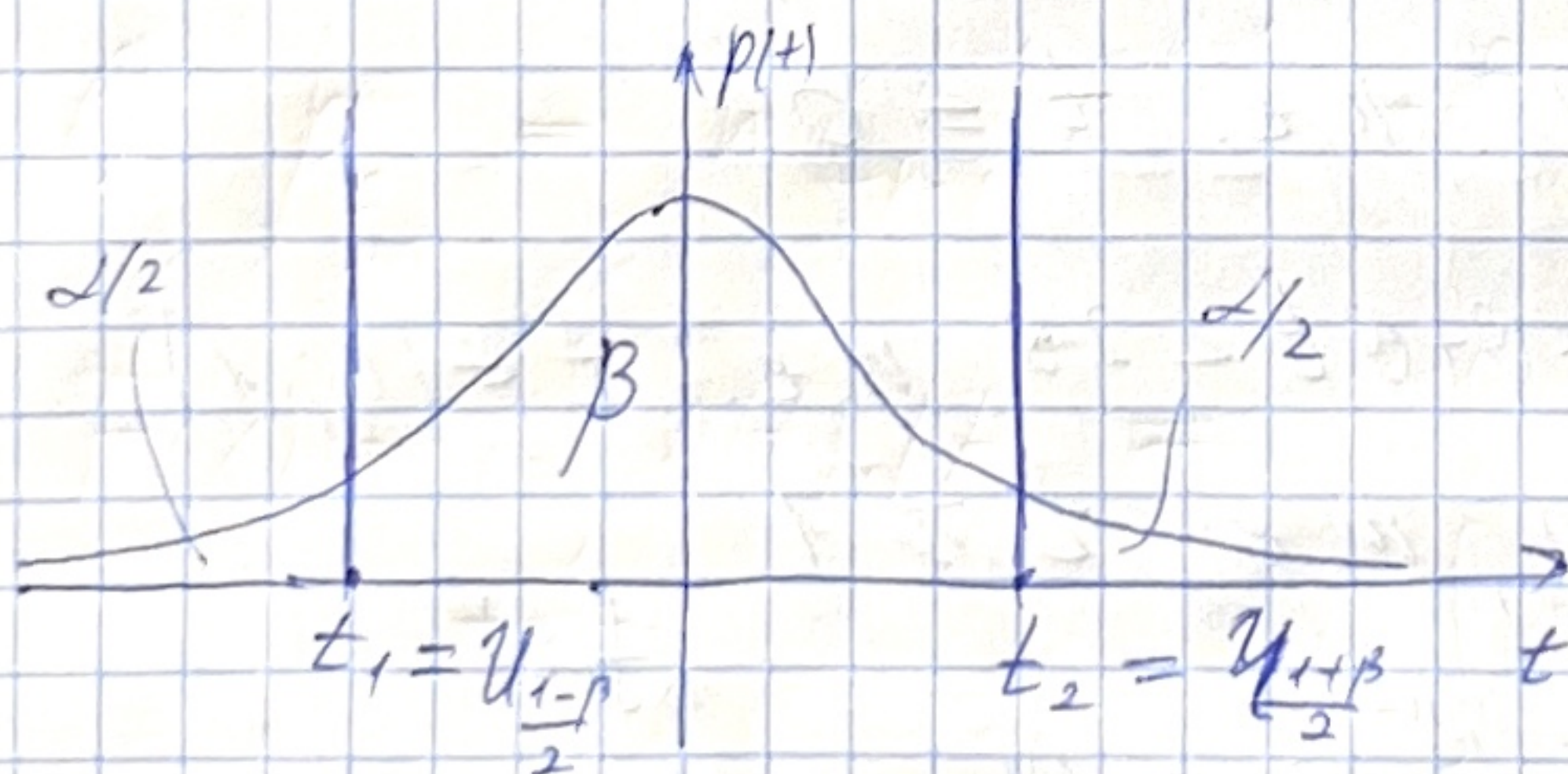
$$\tilde{\theta}_1 = \frac{2}{3} \bar{X} = g(\tilde{\alpha}_1), \quad \sqrt{n} \frac{(\tilde{\theta}_1 - \theta)}{o(\tilde{\theta})} \sim N(0, 1)$$

$$g(\alpha) = \theta, \quad g(\alpha) = \frac{2}{3} \alpha, \quad \nabla g(\alpha) = \frac{2}{3} \neq 0$$

$$\sigma^2(\alpha) = \frac{4}{9} (\alpha_2 - \alpha_1^2)$$

$$\Rightarrow \sqrt{n} \frac{\tilde{\theta}_1 - \theta}{\frac{2}{3} \sqrt{\alpha_2 - \alpha_1^2}} \sim N(0, 1)$$

$$t_1 < \sqrt{n} \frac{\tilde{\theta}_1 - \theta}{\frac{2}{3} \sqrt{\alpha_2 - \alpha_1^2}} < t_2$$



$$\frac{t_1 \cdot \frac{2}{3} \sqrt{\alpha_2 - \alpha_1^2}}{\sqrt{n}} < \tilde{\theta}_1 - \theta < \frac{t_2 \cdot \frac{2}{3} \sqrt{\alpha_2 - \alpha_1^2}}{\sqrt{n}}$$

$$\begin{cases} \theta > \tilde{\theta}_1 - \frac{2 t_2 \sqrt{\alpha_2 - \alpha_1^2}}{3 \sqrt{n}} \\ \theta < \tilde{\theta}_1 - \frac{2 t_1 \sqrt{\alpha_2 - \alpha_1^2}}{3 \sqrt{n}} \end{cases} \quad \text{и.е.}$$

$$\frac{2}{3} \bar{X} - \frac{2 t_2 \sqrt{\alpha_2 - \alpha_1^2}}{3 \sqrt{n}} < \theta < \tilde{\theta}_1 - \frac{2 t_1 \sqrt{\alpha_2 - \alpha_1^2}}{3 \sqrt{n}}$$

$$\left[\frac{2}{3} \bar{X} - \frac{2t_2 \sqrt{2_2 - 2_1^2}}{3\sqrt{n}} < \theta < \frac{2}{3} \bar{X} - \frac{2t_1 \sqrt{2_2 - 2_1^2}}{3\sqrt{n}} \right]$$

$$\text{III. к.} \int_{A(\theta)} \frac{\partial^2}{\partial \theta^2} p(x, \theta) dx + \sum_k \frac{\partial^2}{\partial \theta^2} p_k(\theta) =$$

$$= \left\{ A(\theta) = (\theta; 2\theta), \quad p(x, \theta) = \frac{1}{\theta}, \quad p_k = 0 \right\} =$$

$$\left\{ \frac{\partial^2}{\partial \theta^2} \left(\frac{1}{\theta} \right) = \frac{\partial}{\partial \theta} \left(-\frac{1}{\theta^2} \right) = \frac{2}{\theta^3} \right\} =$$

$$= \int_{\theta}^{\theta} \frac{2}{\theta^3} dx = \frac{2 \cdot \theta}{\theta^3} = \frac{2}{\theta^2} \neq 0 \Rightarrow \text{вероятно}$$

не явл. строго регулярной $\Rightarrow$ проверка
 асимпт. провер. интервала по ДМТ (с помощью
 $\sqrt{n} (f(\hat{\theta}) - f(\theta)) \leadsto N(0, f'(\theta) \cdot \frac{1}{I(\theta)})$) не
 работает.

Сравнение интервалов:

$\theta = 10$

Точный доверительный интервал	: 9.938446 < θ < 10.120453, длина = 0.182007
Асимптотический доверительный интервал	: 9.721661 < θ < 10.496284, длина = 0.774624
Непараметрический бутстрап	: 9.937188 < θ < 10.071854, длина = 0.134666
Параметрический бутстрап	: 9.938458 < θ < 10.116862, длина = 0.178405